Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 10.2 High-Frequency Response of CS and CE Amplifier

Recap

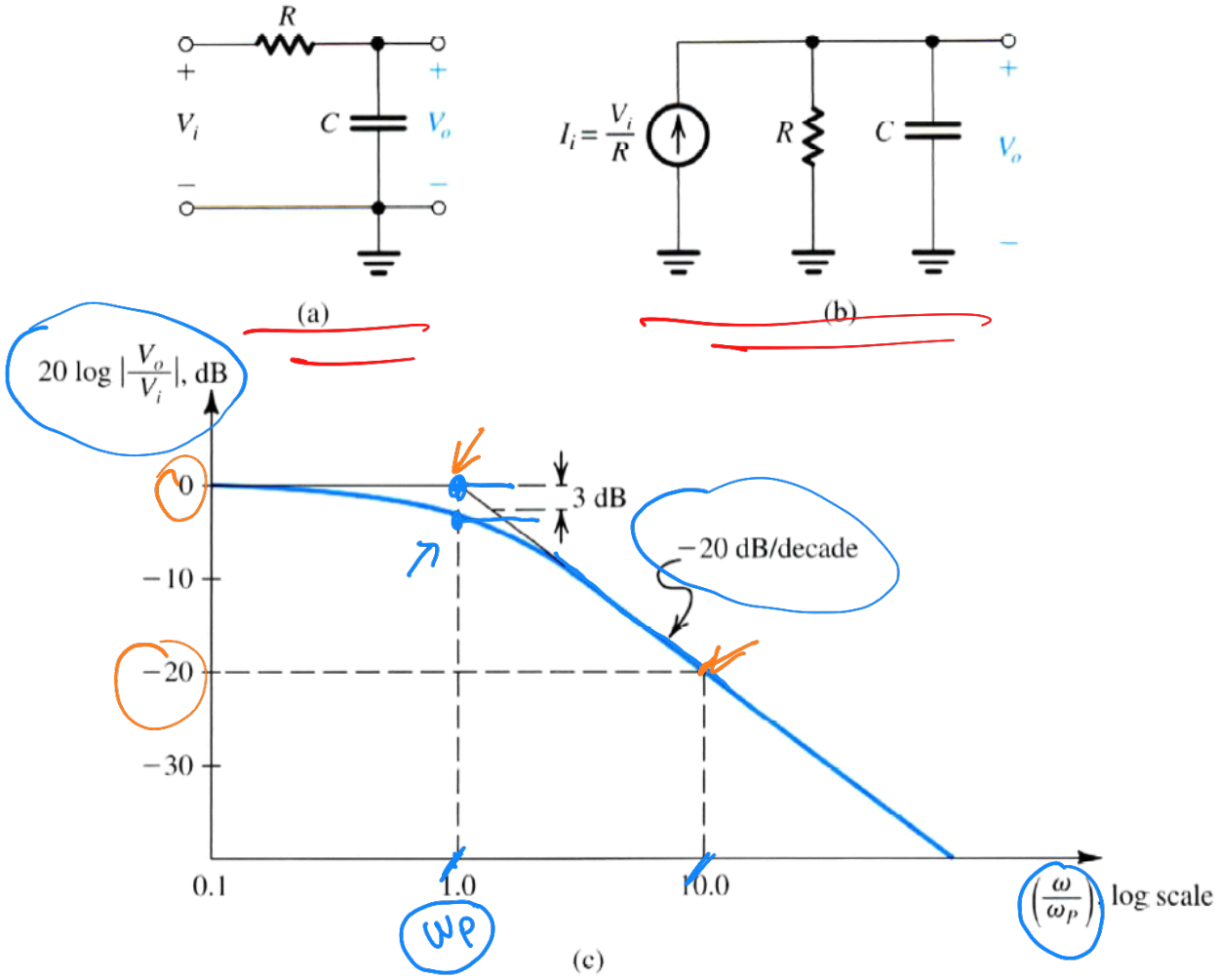


Figure 10.10 The low-pass single-time-constant circuit in its (a) Thévenin form and (b) Norton form. (c) Bode plot for $|V_o/V_i|$.

$$H(s) = \frac{V_o}{V_i} = \frac{1}{1 + \frac{s}{\omega_P}}$$

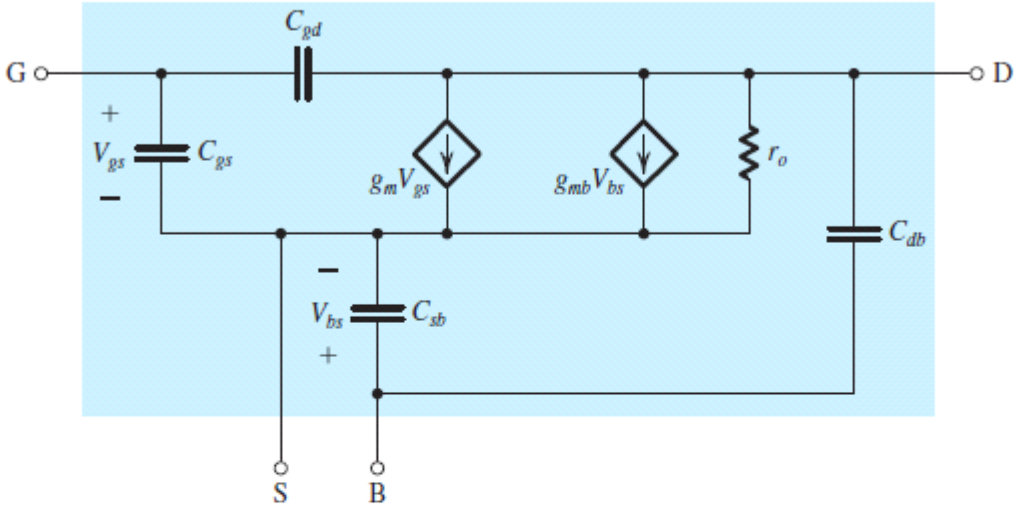
$$\omega_P = \frac{1}{\tau} \qquad \tau = CR$$

$$\left|\frac{V_o}{V_i}\right| = \frac{1}{\sqrt{1 + \left(\frac{\omega}{\omega_P}\right)^2}}$$

Summary Table

Table 10.1 The MOSFET High-Frequency Model

Model



Model Parameters

$$g_m = \mu_n C_{ox} \frac{W}{L} |V_{OV}| = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} = \frac{2I_D}{|V_{OV}|}$$

$$g_{mb} = \chi g_m, \quad \chi = 0.1 \text{ to } 0.2$$

$$r_o = |V_A| / I_D$$

$$C_{gs} = \frac{2}{3} WLC_{ox} + WL_{ov} C_{ox}$$

$$C_{gd} = WL_{ov} C_{ox}$$

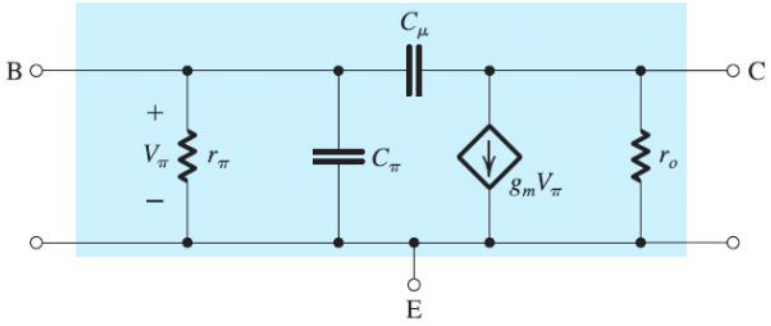
$$C_{sb} = \frac{C_{sb0}}{\sqrt{1 + \frac{|V_{SB}|}{V_0}}}$$

$$C_{db} = \frac{C_{db0}}{\sqrt{1 + \frac{|V_{DB}|}{V_0}}}$$

$$f_T = \frac{g_m}{2\pi (C_{gs} + C_{gd})}$$

Summary Table

Table 10.2 The BJT High-Frequency Model

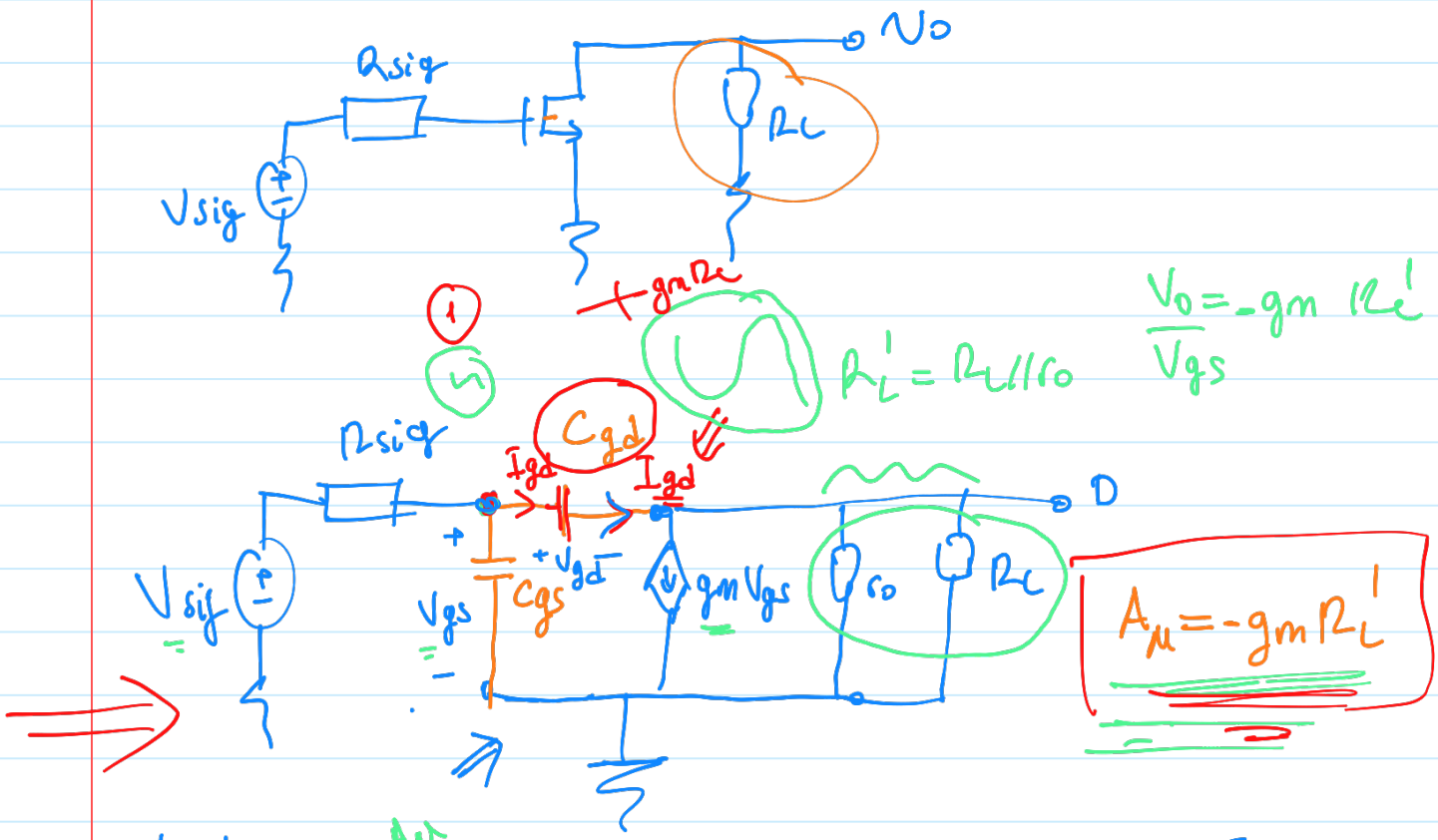


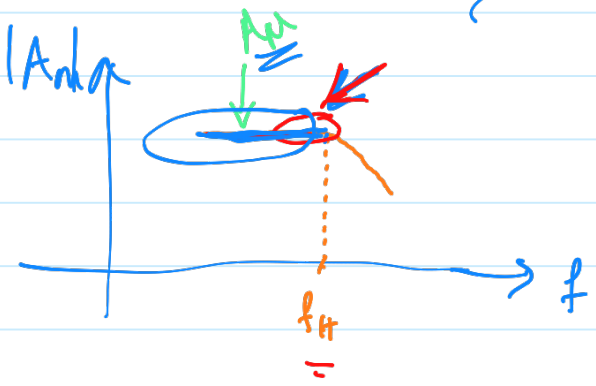
$g_m = I_C/V_T$ $\quad r_o = |V_A|/I_C$ $\quad r_\pi = \beta_0/g_m$

$C_\pi + C_\mu = \dfrac{g_m}{2\pi f_T}$ $\quad C_\pi = C_{de} + C_{je}$ $\quad C_{de} = \tau_F g_m$ $\quad C_{je} \simeq 2C_{je0}$

$C_\mu = C_{jc0} \Big/ \left(1 + \dfrac{|V_{CB}|}{V_{0c}}\right)^m$ $\quad m = 0.3\text{–}0.5$

The Common Source Amplifier





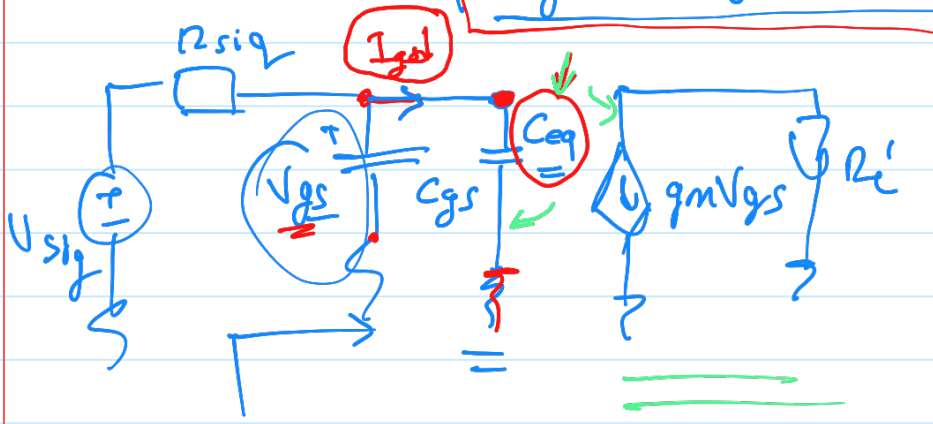
$g_m V_{gs} \gg I_{gd}$

$V_o \simeq -(g_m V_{gs}) R_L'$

$V_o = -g_m R_L' V_{gs}$

$I_{gd} = sC_{gd} V_{gd} = sC_{gd}(V_{gs} - V_o)$

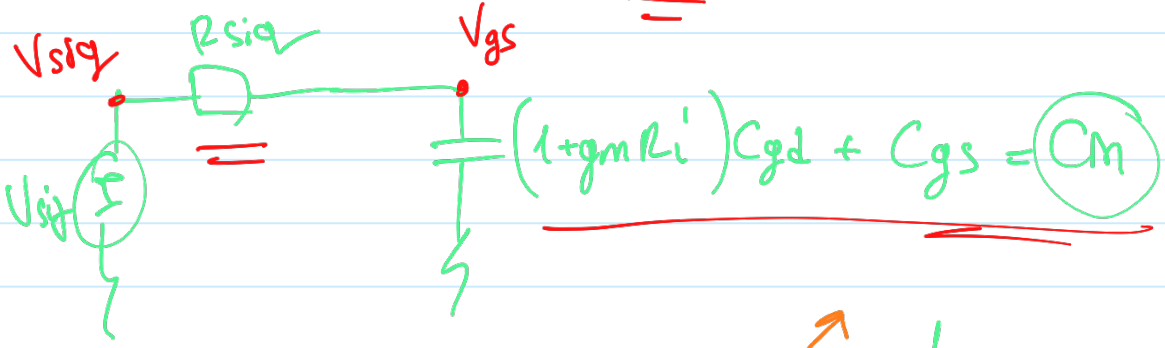
$$I_{gd} = sC_{gd}(1 + g_m R_L') V_{gs}$$



$$= \begin{cases} I_{gd} = s C_{eq} V_{gs} \\ I_{gd} = sC_{gd}(1+g_m R_i') V_{gs} \end{cases}$$

$$C_{eq} = (1+g_m R_i') C_{gd}$$

A_M Miller Effect



$$\frac{V_{gs}}{V_{sig}} = \frac{1}{1+\frac{s}{\omega_p}}$$

$$\omega_p = \frac{1}{R_{sig} C_{in}}$$

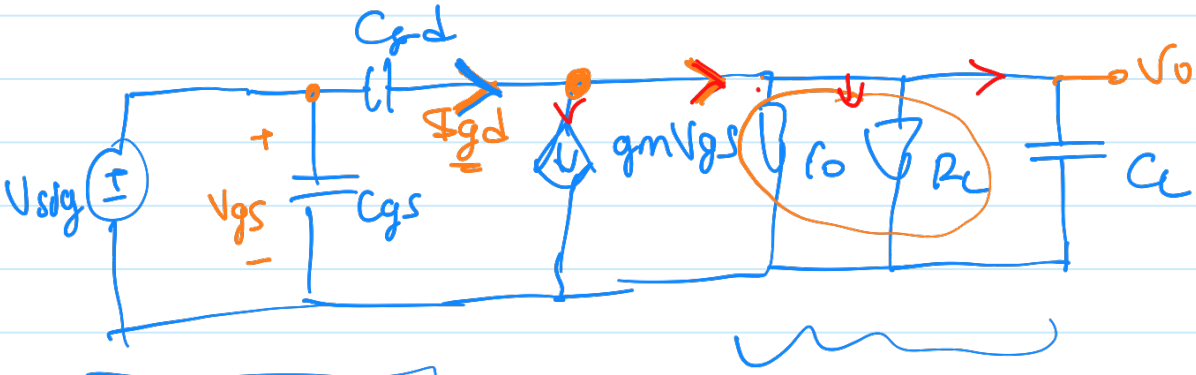
$$\frac{V_o}{V_{sig}} = \frac{-g_m R_L'}{1+\frac{s}{\omega_p}} = \frac{A_M}{1+\frac{s}{\omega_H}}$$

$$f_H = \frac{\omega_H}{2\pi} = \frac{1}{2\pi C_{in} R_{sig}}$$

R_{sig} is large



The Common Source Amplifier (Rsig is Low)



$$V_{gs} = V_{sig}$$

$$I_{gd} = s C_{gd} (V_{gs} - V_o)$$

$$I_{gd} = g_m V_{gs} + \frac{V_o}{R_L'} + s C_L V_o$$

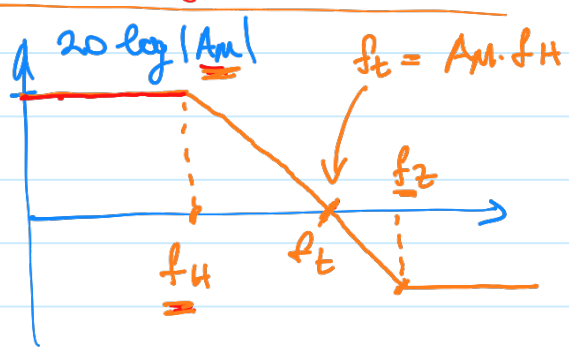
$$H(s) = \frac{V_o}{V_{sig}} = -g_m R_L' \frac{1 - s(C_{gd}/g_m)}{1 + s(C_L + C_{gd}) R_L'}$$

$$f_z = \frac{g_m}{2\pi C_{gd}}$$

$$f_H = \frac{1}{2\pi (C_L + C_{gd}) R_L'}$$

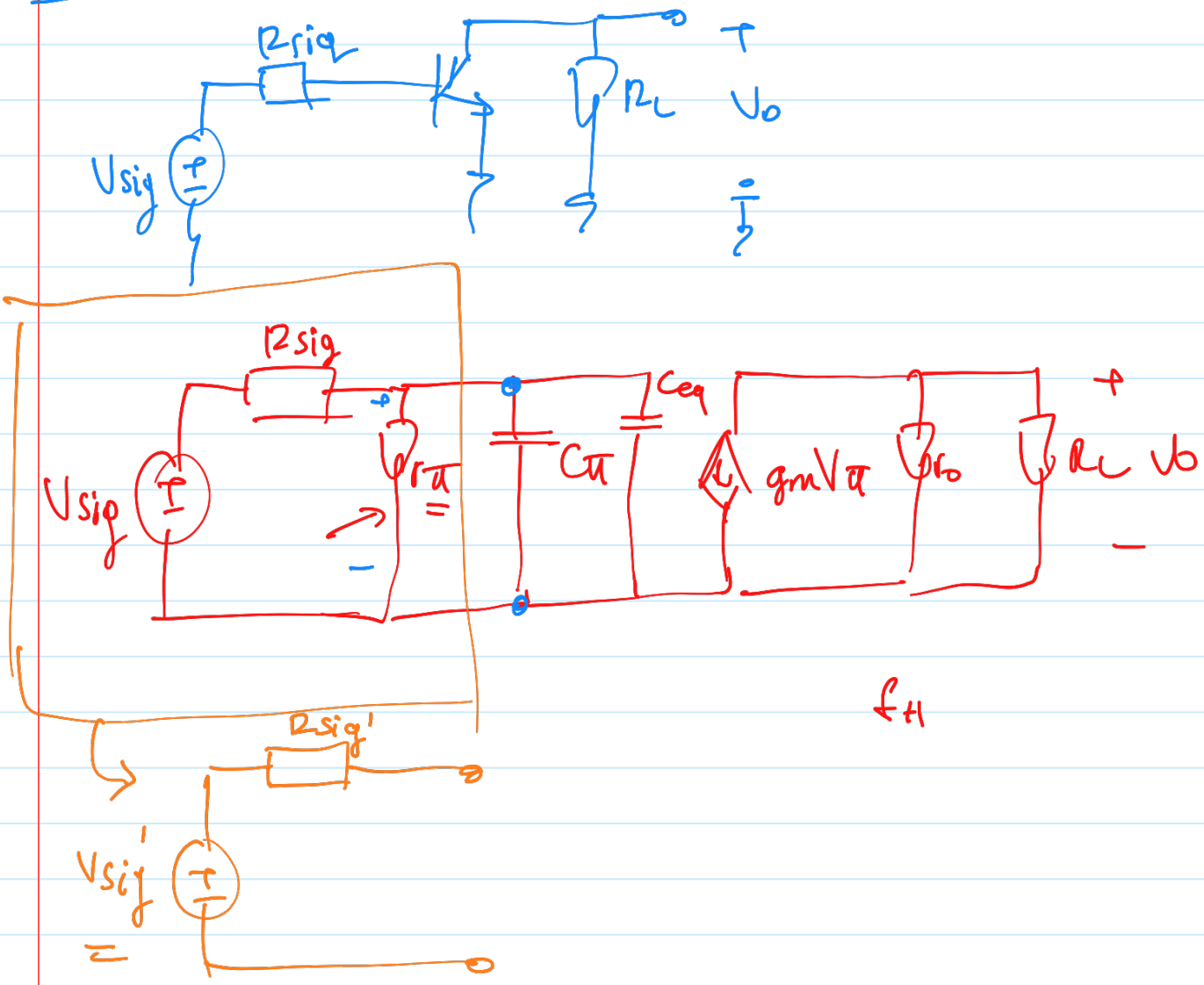
$$\frac{f_z}{f_H} = (g_m R_L') \left(1 + \frac{C_L}{C_{gd}}\right)$$

$f_z \gg f_H$



$$f_t = \frac{g_m}{2\pi (C_L + C_{gd})}$$

The Common Emitter Amplifier



$$V_{sig}' = V_{sig}\frac{r_\pi}{r_\pi + R_{sig}} \qquad R_{sig}' = R_{sig} \,//\, r_\pi$$

$$f_H = \frac{1}{2\pi C_{in} R'_{sig}}$$

$$C_{in} = C_\pi + C_\mu\left(1 + g_m R_L'\right)$$

Template